

IDEABATIC SMILE GO Project Uniform Cooling Solutions

By Karen, Kerry and Kavita

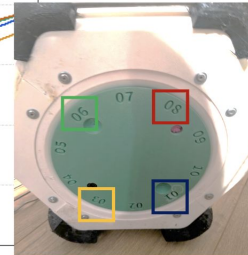
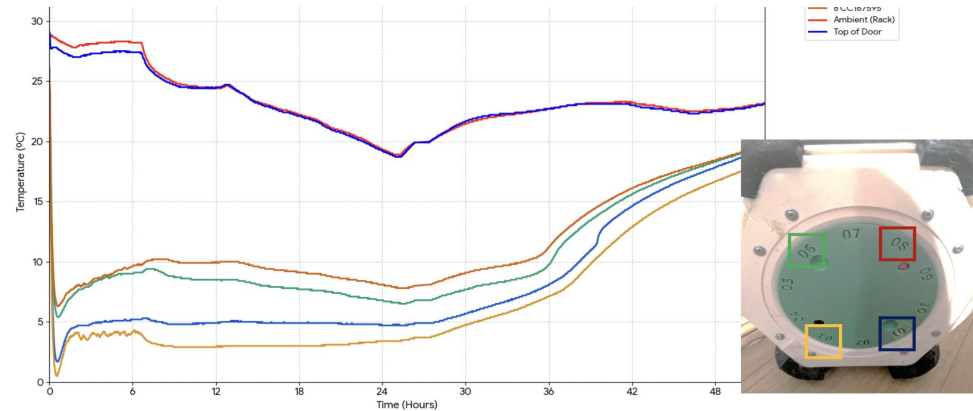
Working with Kitty and Oliver



Problem Overview

So Our Project Focus is

SMILE GO - Investigating the causes of non-uniform cooling & potential ways to reduce it



Solution Overview

Tests Conducted

Control Tests

Door Effect Test

Insert Test

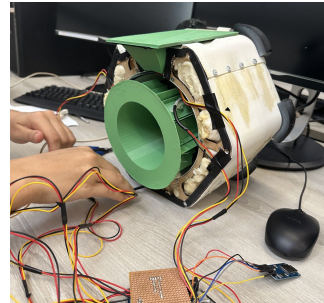
Improved Contact Test

Convection Test (Radial Planes)

Combination Test 1 - Radial Planes and Improved Contact

Combination Test 2 - Radial Planes, Improved Contact, and Botte Insert

1. Identify possible causes of asymmetry.
2. Design and perform tests.
3. Analyse results and identify promising causes.
4. Offer design solutions and further testing to be conducted.



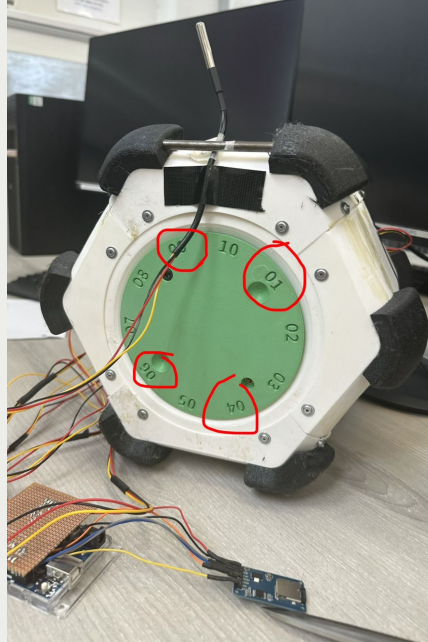
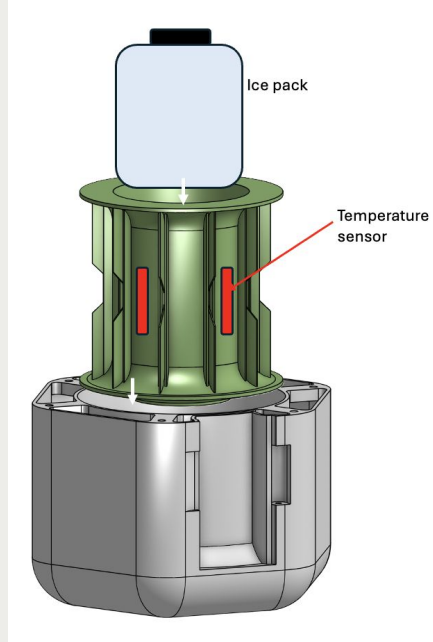
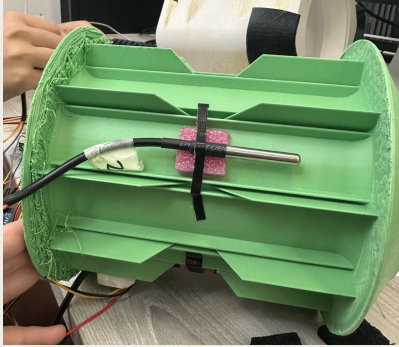
Benefits



- Helps to give life-saving vaccines to remote areas
- Medication remains safe and effective for administration for longer
- Supports reduction of vaccine wastage
- Affordable and portable box
- Collected data to support future design improvements
- Aligns with UN sustainable goals

Tests

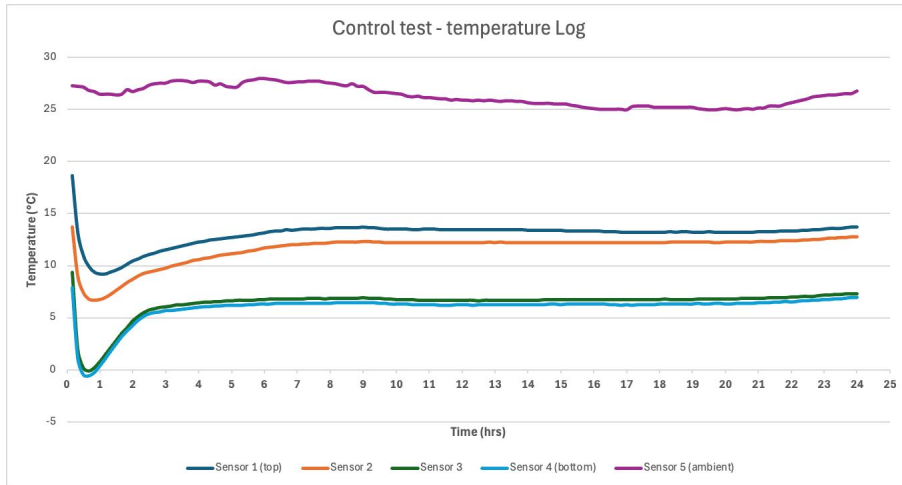
Experimental set-up



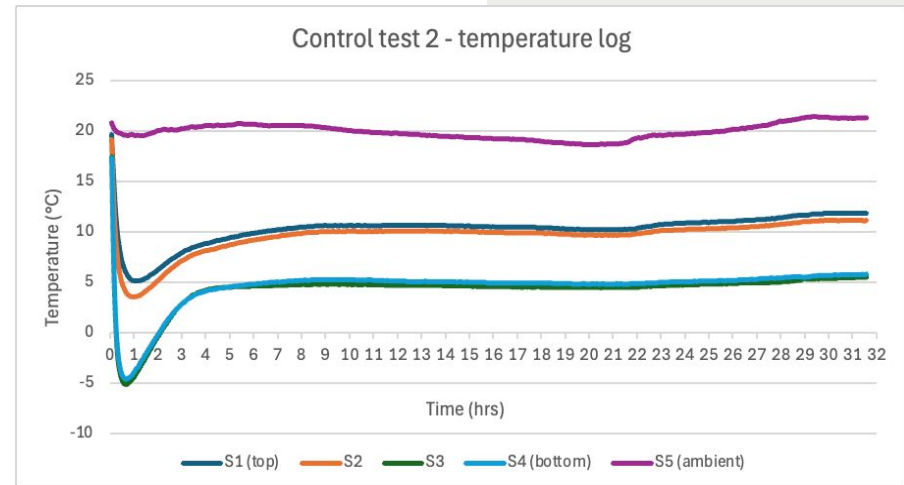
Control Test 1 & 2

01

Control test 1 (during Heat wave):



Control test 2 (after Heat wave):



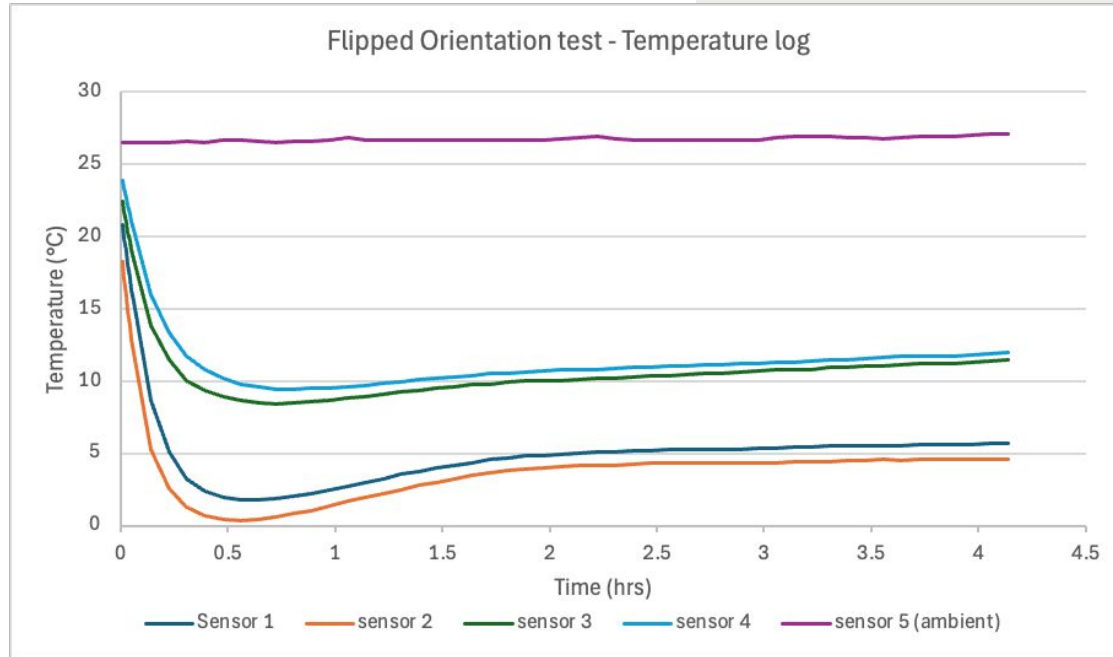
Results

Temp difference: $\sim 7^{\circ}\text{C}$ for control test 1 and $\sim 6^{\circ}\text{C}$ for control test 2

Door Test

02

Cooler oriented upside down with the door on the bottom to test its effects



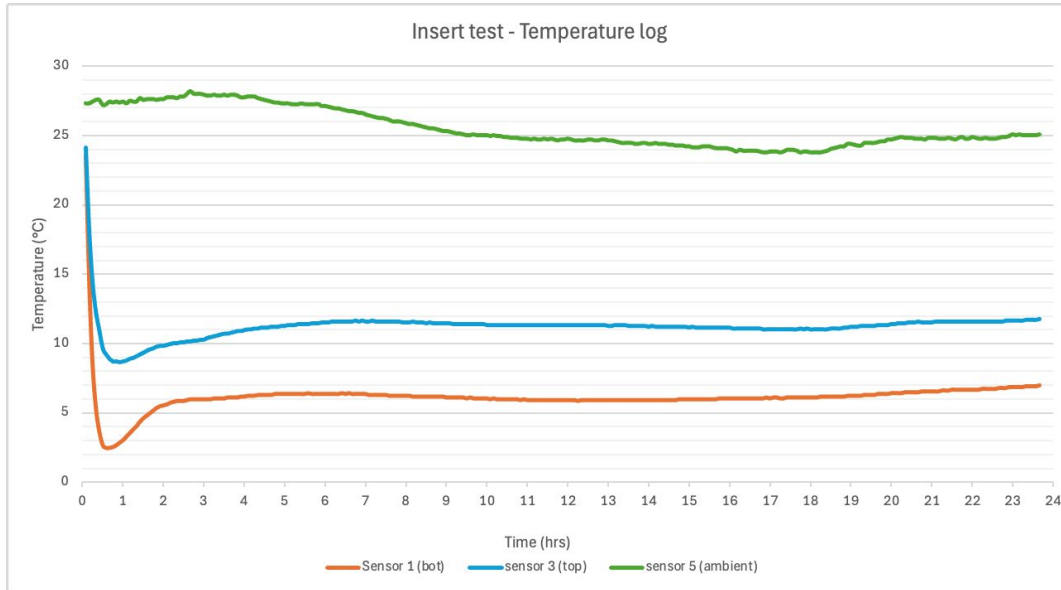
Results

No visible changes

Insert Test

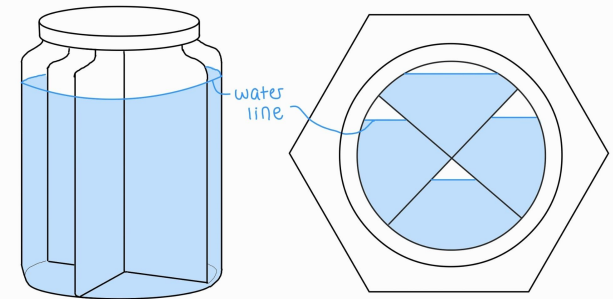
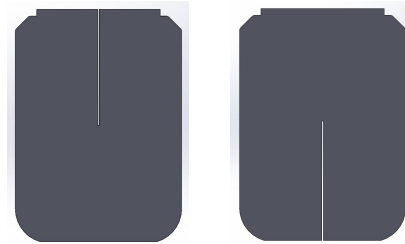
03

3D printed inserts hot glued into the ice pack bottle before placing it into the carousel.



At 12 hours:

- Insert test: 5.3°C
- Control test: 7.2°C



Results

- 2 degree improvement in the temperature difference seen

Material Research On Inserts

Materials & Manufacturing

- PLA for rapid, low-cost prototyping
- Hot glue used for temporary sealing

Key Findings

- Print porosity caused water permeation
- PLA durability concerns especially under repeated freeze-thaw cycles

Future Improvements

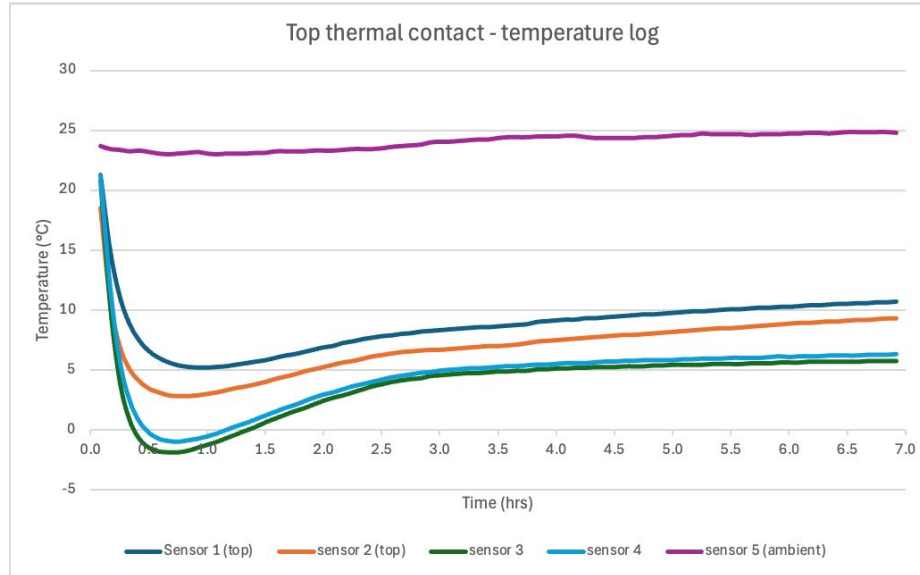
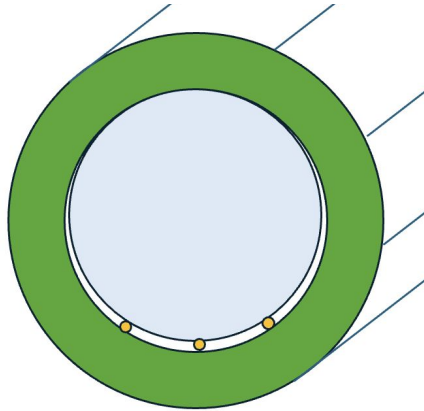
- PETG for improved water resistance
- Integrated blow-moulded chambers for production
- Better coatings/sealants



Improved Contact Test

04

Small inserts attached to bottom of central chamber such that ice pack is in contact with top and raised off of bottom.



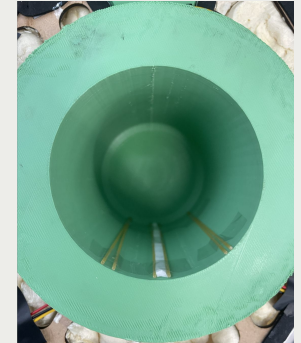
Results

- Reduction in temperature gradient but significant non-uniformity remains

At 7 hrs:

- Top contact test: 4.9°C
- Control 1 test: 7°C
- Control 2 test: 5.5°C

Ambient temp between Control 1 & 2 temp



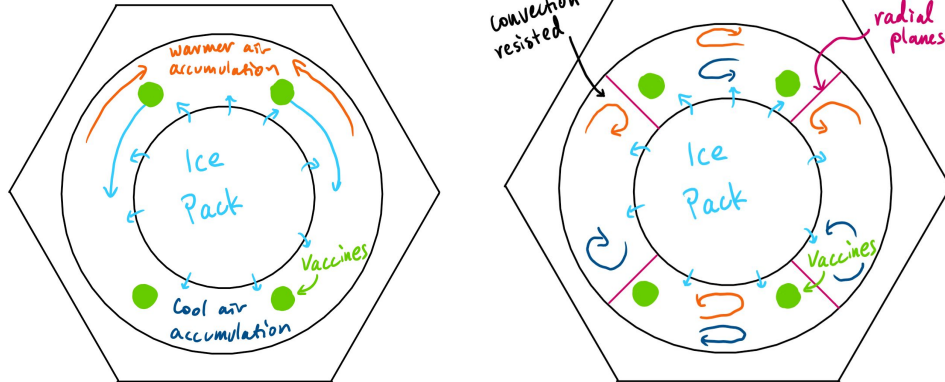
Convection - Radial Planes Test

05

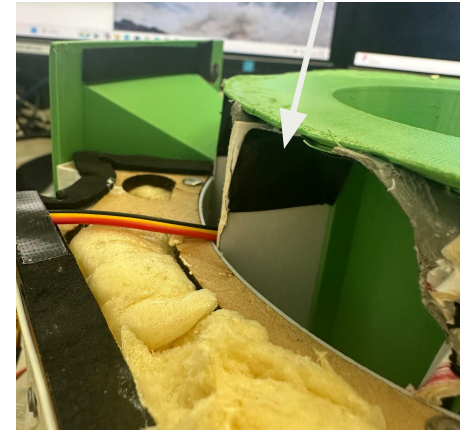
Natural Convection - The separation of hot air and cold air due to hot air rising.

Radial Planes - Stop (or slow down) the cold air from falling all the way to the bottom.

Radial planes did slow down the motion of the cool air falling.



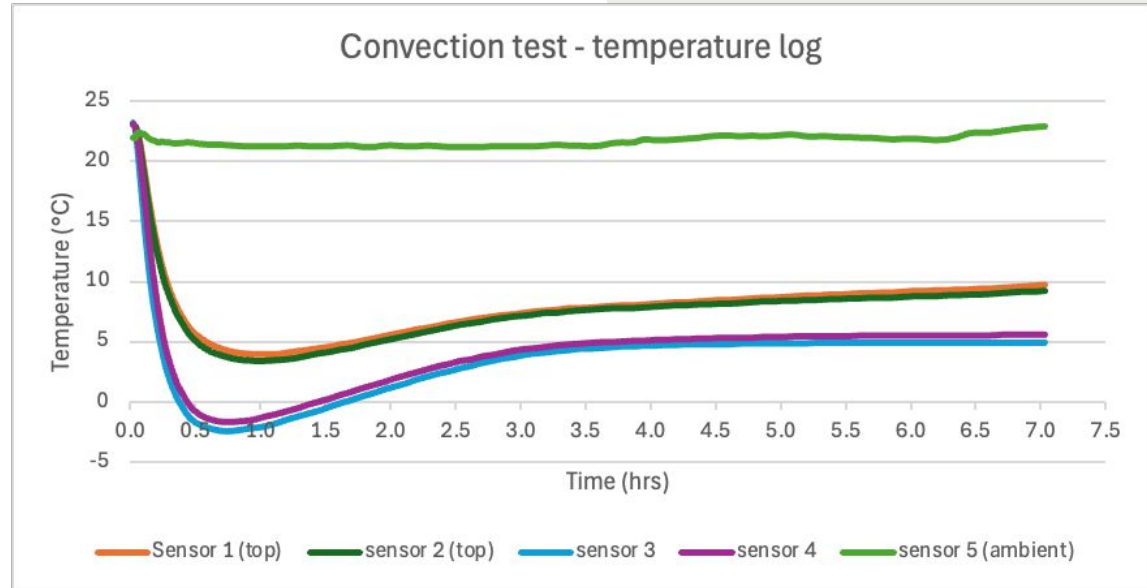
Radial Plane



Convection - Radial Planes Test

06

Results:
Temperature difference ~4C
(Control 2 ~5C)



Radial Planes + Improved Contact (Combination 1) 07

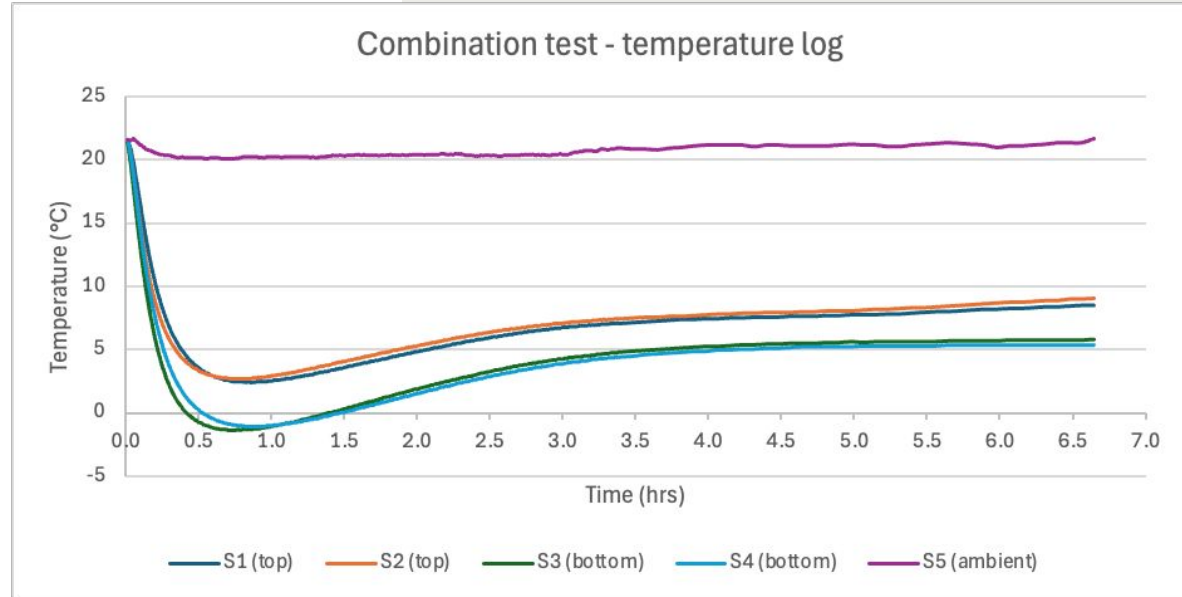
Cool air is generated nearer top section -> its convection is then resisted by radial planes..

Combine factors and observe their interactions.

Results:

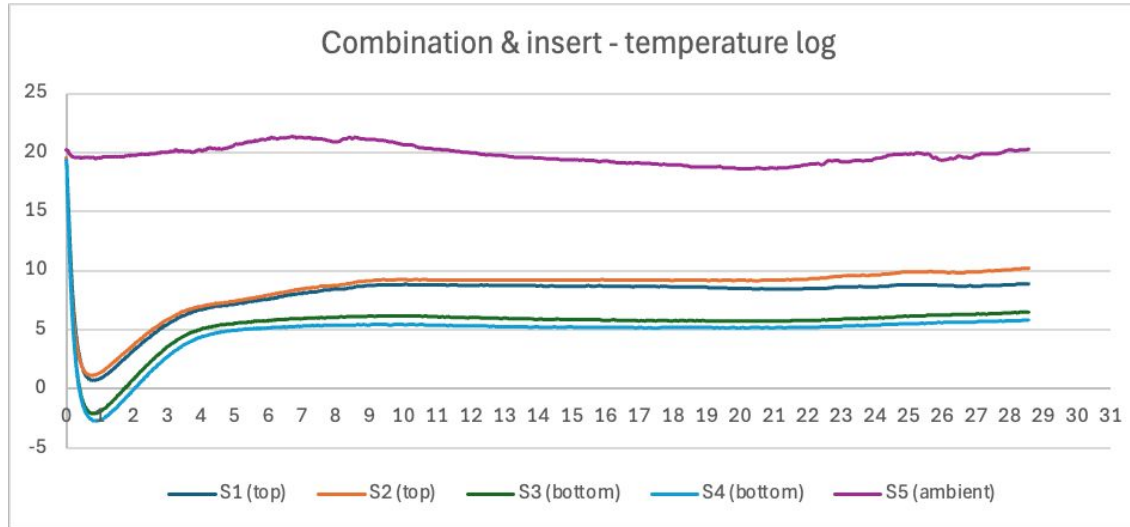
A small improvement was observed - temperature difference ~3.5C

(Compared to control 2 ~5C)



Radial Planes + Improved Contact + Insert Test 08 (Combination 2)

Combining factors in the prior test with bottle inserts included



At 6.5 hours:

- Combo 2 test: 3°C
- Control 2: 5°C

At 24 hours:

- Combo 2 test: 4°C
- Control 2: 5°C

Results

- Slight improvements observed

Fan Test

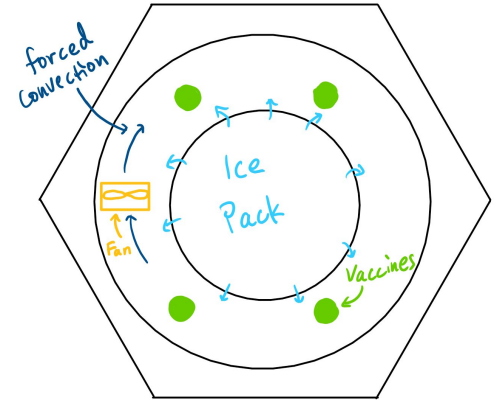
09

Forced Convection - Use small fans to mix the hot and cold air within the vaccine chamber.

Mixing the air in the chamber does not help narrow down the **source** of the non-uniform cooling.

This is a mitigation strategy - a possible solution.

25mm x 25mm x 10mm
USB fan

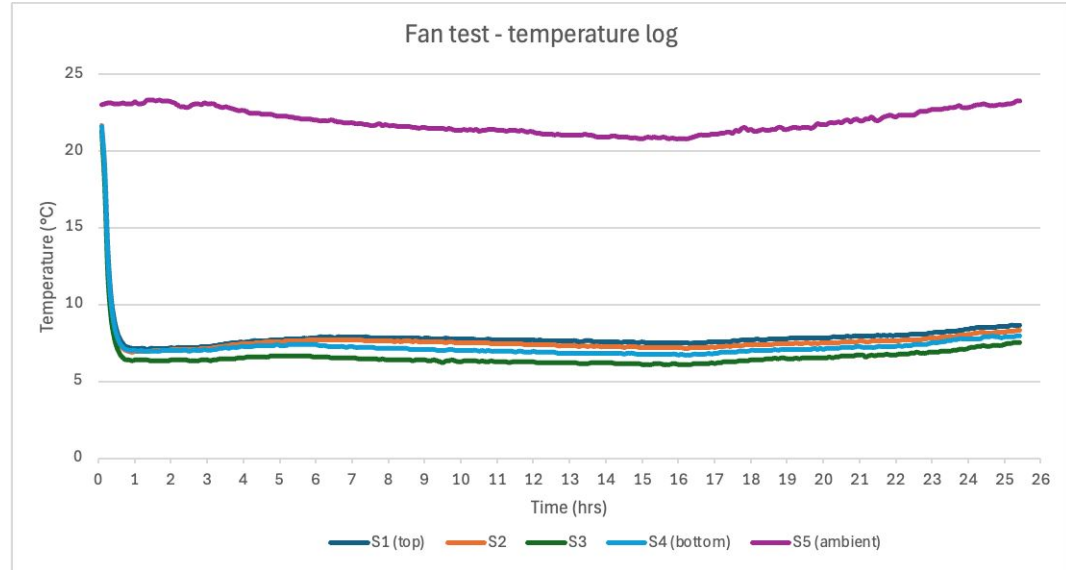


Fan Test

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Caveats of using a fan:

- Requires power source
- Generates small amount of electrical heat



Conclusions

Findings

Overall Findings

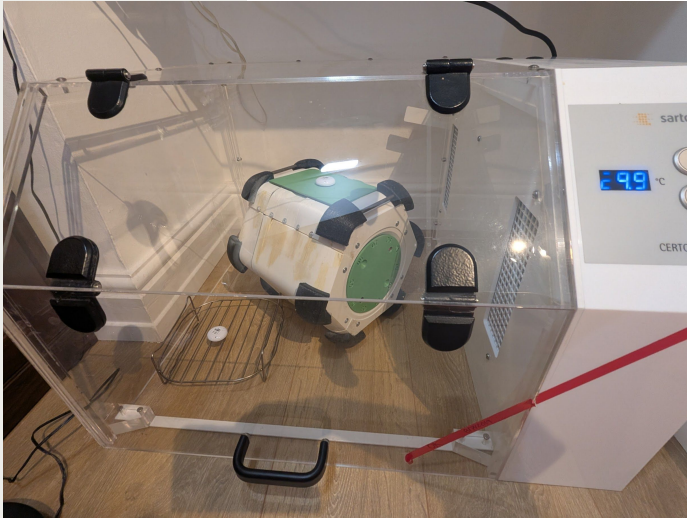
- Cooling remains non-uniform throughout the carrier in all tests
- Current insulation is insufficient to maintain the required 2-8°C range
- Temperatures fall below 2°C during the early stages of cooling
- Risk of lower vaccines freezing
- Risk of upper vaccines expiring

Summary of results

- Door test was ineffective
- Insert, improved contact and convection tests showed improvements
- A number of factors contributed to the asymmetrical cooling
- Fans within the carousel causes uniformity

Potential Further Testing

11

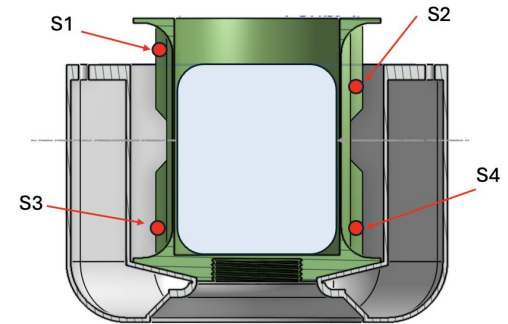


- More rigorous testing with a larger number of tests
- Controlled ambient environment
- Longer test cycle length
- Ice pack insert design variations
- Fully airtight radial planes/convection test
- Assess impact of wear of SMILE Go
- Chamber temperature vs vaccine temperature

Potential Design Solutions

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- Insulation
 - **More uniform & better quality**
 - Thicker insulation on top side
 - Different insulation material on top and bottom
- Ice Pack
 - Use 2: additional one at top
 - Identically dimensioned casing
- Carousel
 - **Add radial baffles**
 - Fill air gaps around vials (foam)
- Orientation
 - Operate the device vertically
- Active solution e.g. fan



Vertical test: $\sim 1^{\circ}\text{C}$ between
S2 & S4

Thanks for listening